

Experiment 2 — Data Visualization

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Aim

To create meaningful visualizations using bar charts and box plots on the Aadhar Enrollment dataset to represent patterns, trends, and distributions across states, districts, and age groups, and to interpret the insights derived from them.

Theory

Data Visualization is the graphical representation of data and information. It transforms raw numbers into visual formats such as charts, graphs, and plots, making it easier to identify patterns, trends, outliers, and relationships that would otherwise be difficult to detect in tabular form.

1. Importance of Visualization in Data Science: After EDA, visualization serves as the next critical step. It enables analysts to communicate findings effectively, validate assumptions, and guide decisions. Python's matplotlib and seaborn libraries are widely used for this purpose.

2. Bar Chart: A bar chart represents categorical data with rectangular bars. The height or length of each bar is proportional to the value it represents. Bar charts are ideal for comparing quantities across categories.

- **Grouped Bar Chart** — Displays multiple sub-categories side by side for each main category. Used here to compare age group enrollments (0–5, 5–17, 18+) across different states.
- **Stacked Bar Chart** — Layers sub-categories on top of each other within a single bar, showing both the total and the contribution of each part. Used here to show combined enrollment per state.
- **Simple Bar Chart** — Used to display the Top 10 districts by total enrollment, with value labels on each bar for easy reading.

3. Box Plot: A box plot (also called a box-and-whisker plot) summarizes the distribution of a dataset through five statistics:

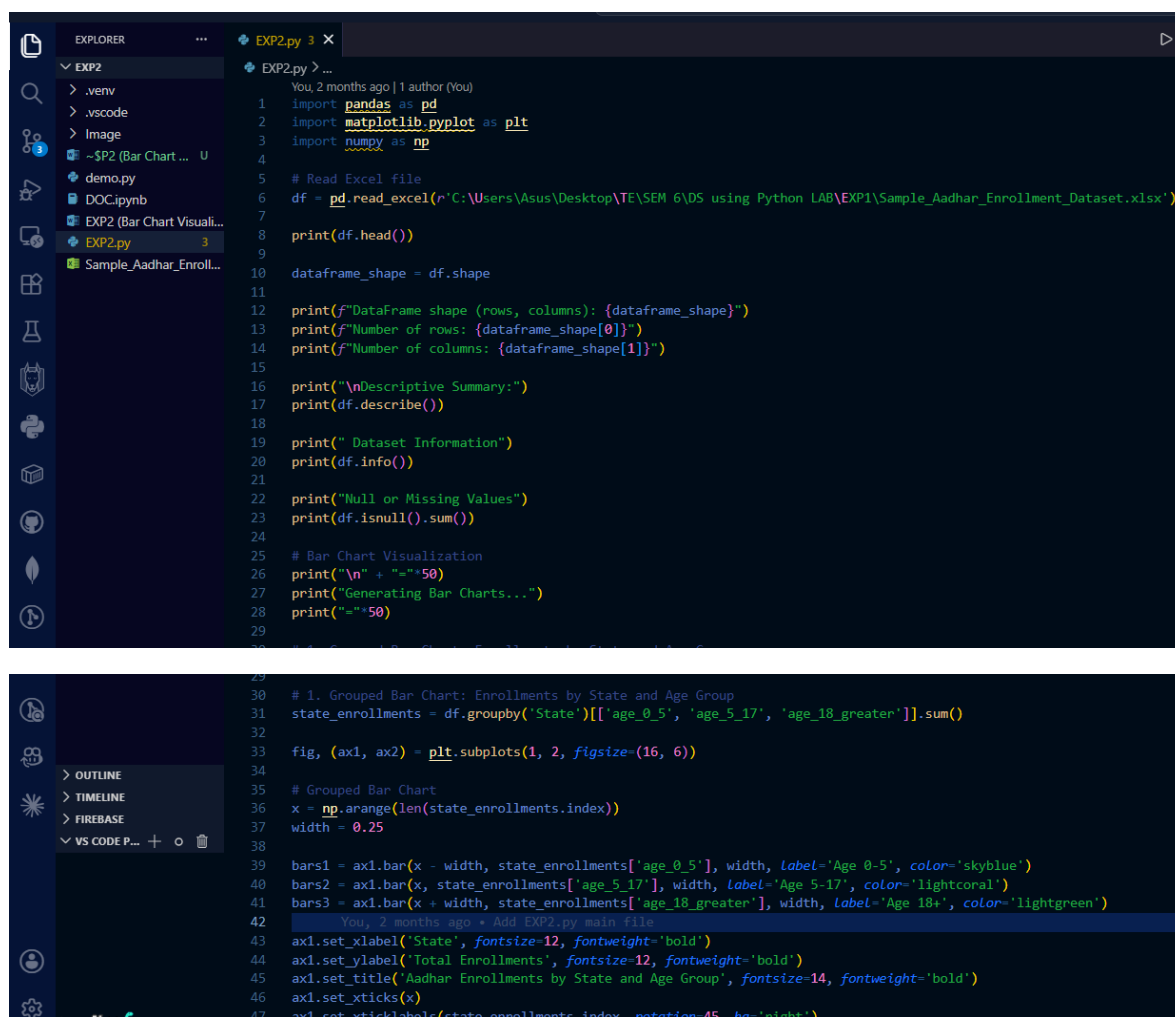
- **Minimum and Maximum** — The ends of the whiskers (excluding outliers).
- **Q1 (25th percentile)** — Lower edge of the box.
- **Median (Q2 / 50th percentile)** — Line inside the box.
- **Q3 (75th percentile)** — Upper edge of the box.
- **IQR (Interquartile Range)** — $Q3 - Q1$, representing the middle 50% of data.
- **Outliers** — Points beyond $1.5 \times \text{IQR}$ from Q1 or Q3, plotted as individual dots.

Box plots are ideal for comparing distributions across multiple groups and identifying skewness and outliers.

4. Data Aggregation Before Plotting: Before plotting, raw data is aggregated meaningfully:

- A new column Total_Enrollment is computed by summing the three age-group columns (age_0_5, age_5_17, age_18_greater), handling NaN values gracefully.
- Data is grouped by State (summing age-group columns) to compare state-wise enrollment.
- Data is grouped by district (summing Total_Enrollment) and sorted to extract the Top 10 districts.

Execution

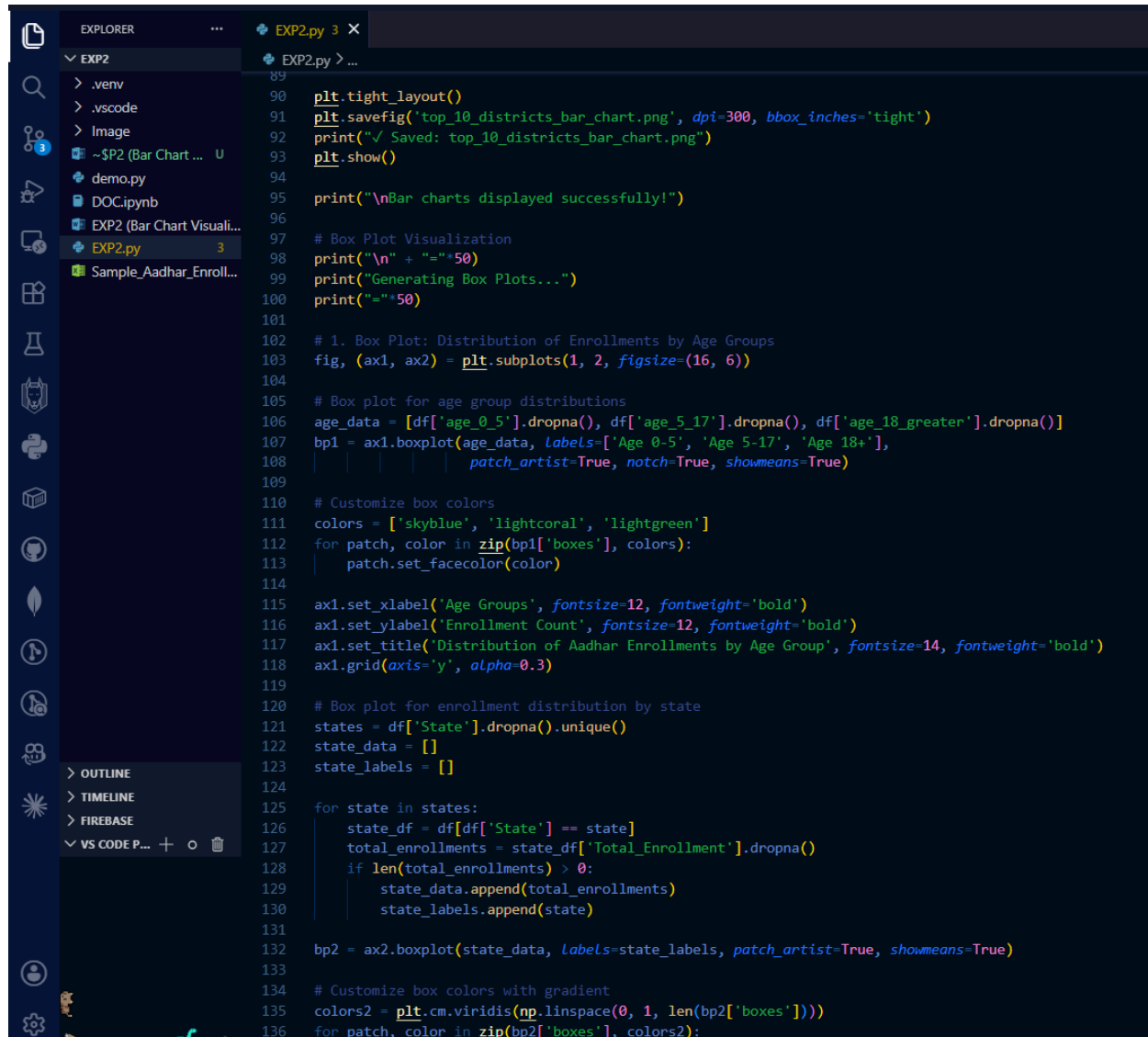


```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3 import numpy as np
4
5 # Read Excel file
6 df = pd.read_excel(r"C:\Users\Asus\Desktop\TE\SEM 6\DS using Python LAB\EXP1\Sample_Aadhar_Enrollment_Dataset.xlsx')
7
8 print(df.head())
9
10 dataframe_shape = df.shape
11
12 print(f"DataFrame shape (rows, columns): {dataframe_shape}")
13 print(f"Number of rows: {dataframe_shape[0]}")
14 print(f"Number of columns: {dataframe_shape[1]}")
15
16 print("\nDescriptive Summary:")
17 print(df.describe())
18
19 print("Dataset Information")
20 print(df.info())
21
22 print("Null or Missing Values")
23 print(df.isnull().sum())
24
25 # Bar Chart Visualization
26 print("\n" + "="*50)
27 print("Generating Bar Charts...")
28 print("="*50)
29
30 # 1. Grouped Bar Chart: Enrollments by State and Age Group
31 state_enrollments = df.groupby('State')[['age_0_5', 'age_5_17', 'age_18_greater']].sum()
32
33 fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(16, 6))
34
35 # Grouped Bar Chart
36 x = np.arange(len(state_enrollments.index))
37 width = 0.25
38
39 bars1 = ax1.bar(x - width, state_enrollments['age_0_5'], width, label='Age 0-5', color='skyblue')
40 bars2 = ax1.bar(x, state_enrollments['age_5_17'], width, label='Age 5-17', color='lightcoral')
41 bars3 = ax1.bar(x + width, state_enrollments['age_18_greater'], width, label='Age 18+', color='lightgreen')
42
43 ax1.set_xlabel('State', fontsize=12, fontweight='bold')
44 ax1.set_ylabel('Total Enrollments', fontsize=12, fontweight='bold')
45 ax1.set_title('Aadhar Enrollments by State and Age Group', fontsize=14, fontweight='bold')
46 ax1.set_xticks(x)
47 ax1.set_xticklabels(state_enrollments.index, rotation=45, ha='right')
```

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43 ax1.set_xlabel('State', fontsize=12, fontweight='bold')
44 ax1.set_ylabel('Total Enrollments', fontsize=12, fontweight='bold')
45 ax1.set_title('Aadhar Enrollments by State and Age Group', fontsize=14, fontweight='bold')
46 ax1.set_xticks(x)
47 ax1.set_xticklabels(state_enrollments.index, rotation=45, ha='right')
48 ax1.legend()
49 ax1.grid(axis='y', alpha=0.3)
50
51 # Stacked Bar Chart
52 ax2.bar(state_enrollments.index, state_enrollments['age_0_5'], label='Age 0-5', color='skyblue')
53 ax2.bar(state_enrollments.index, state_enrollments['age_5_17'], bottom=state_enrollments['age_0_5'],
54         label='Age 5-17', color='lightcoral')
55 ax2.bar(state_enrollments.index, state_enrollments['age_18_greater'],
56         bottom=state_enrollments['age_0_5'] + state_enrollments['age_5_17'],
57         label='Age 18+', color='lightgreen')
58
59 ax2.set_xlabel('State', fontsize=12, fontweight='bold')
60 ax2.set_ylabel('Total Enrollments', fontsize=12, fontweight='bold')
61 ax2.set_title('Stacked Aadhar Enrollments by State', fontsize=14, fontweight='bold')
62 ax2.set_xticklabels(state_enrollments.index, rotation=45, ha='right')
63 ax2.legend()
64 ax2.grid(axis='y', alpha=0.3)
65
66 plt.tight_layout()
67 plt.savefig('bar_charts_by_state.png', dpi=300, bbox_inches='tight')
68 print("✓ Saved: bar_charts_by_state.png")
69 plt.show()
70
71 # 2. Simple Bar Chart: Top 10 Districts by Total Enrollment
72 df['Total_Enrollment'] = df['age_0_5'].fillna(0) + df['age_5_17'].fillna(0) + df['age_18_greater'].fillna(0)
73 top_districts = df.groupby('district')['Total_Enrollment'].sum().sort_values(ascending=False).head(10)
74
75 plt.figure(figsize=(12, 6))
76 bars = plt.bar(range(len(top_districts)), top_districts.values, color='steelblue', edgecolor='black')
77 plt.xlabel('District', fontsize=12, fontweight='bold')
78 plt.ylabel('Total Enrollments', fontsize=12, fontweight='bold')
79 plt.title('Top 10 Districts by Total Aadhar Enrollments', fontsize=14, fontweight='bold')
80 plt.xticks(range(len(top_districts)), top_districts.index, rotation=45, ha='right')
81 plt.grid(axis='y', alpha=0.3)
82
83 # Add value labels on bars
84 for i, bar in enumerate(bars):
85     height = bar.get_height()
86     plt.text(bar.get_x() + bar.get_width()/2., height,
87             f'{int(height)}',
88             ha='center', va='bottom', fontsize=10, fontweight='bold')
89

```



```

89
90 plt.tight_layout()
91 plt.savefig('top_10_districts_bar_chart.png', dpi=300, bbox_inches='tight')
92 print("\n Saved: top_10_districts_bar_chart.png")
93 plt.show()
94
95 print("\nBar charts displayed successfully!")
96
97 # Box Plot Visualization
98 print("\n" + "="*50)
99 print("Generating Box Plots...")
100 print("="*50)
101
102 # 1. Box Plot: Distribution of Enrollments by Age Groups
103 fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(16, 6))
104
105 # Box plot for age group distributions
106 age_data = [df['age_0_5'].dropna(), df['age_5_17'].dropna(), df['age_18_greater'].dropna()]
107 bp1 = ax1.boxplot(age_data, labels=['Age 0-5', 'Age 5-17', 'Age 18+'],
108                 patch_artist=True, notch=True, showmeans=True)
109
110 # Customize box colors
111 colors = ['skyblue', 'lightcoral', 'lightgreen']
112 for patch, color in zip(bp1['boxes'], colors):
113     patch.set_facecolor(color)
114
115 ax1.set_xlabel('Age Groups', fontsize=12, fontweight='bold')
116 ax1.set_ylabel('Enrollment Count', fontsize=12, fontweight='bold')
117 ax1.set_title('Distribution of Aadhar Enrollments by Age Group', fontsize=14, fontweight='bold')
118 ax1.grid(axis='y', alpha=0.3)
119
120 # Box plot for enrollment distribution by state
121 states = df['State'].dropna().unique()
122 state_data = []
123 state_labels = []
124
125 for state in states:
126     state_df = df[df['State'] == state]
127     total_enrollments = state_df['Total_Enrollment'].dropna()
128     if len(total_enrollments) > 0:
129         state_data.append(total_enrollments)
130         state_labels.append(state)
131
132 bp2 = ax2.boxplot(state_data, labels=state_labels, patch_artist=True, showmeans=True)
133
134 # Customize box colors with gradient
135 colors2 = plt.cm.viridis(np.linspace(0, 1, len(bp2['boxes'])))
136 for patch, color in zip(bp2['boxes'], colors2):

```

```

EXPLORER
EXP2
  .venv
  .vscode
  Image
  ~$P2 (Bar Chart ... U
  demo.py
  DOC.ipynb
  EXP2 (Bar Chart Visuali...
  EXP2.py 3
  Sample_Aadhar_Enroll...

EXP2.py > ...
133
134 # Customize box colors with gradient
135 colors2 = plt.cm.viridis(np.linspace(0, 1, len(bp2['boxes'])))
136 for patch, color in zip(bp2['boxes'], colors2):
137     patch.set_facecolor(color)
138
139 ax2.set_xlabel('State', fontsize=12, fontweight='bold')
140 ax2.set_ylabel('Total Enrollment', fontsize=12, fontweight='bold')
141 ax2.set_title('Distribution of Total Enrollments by State', fontsize=14, fontweight='bold')
142 ax2.set_xticklabels(state_labels, rotation=45, ha='right')
143 ax2.grid(axis='y', alpha=0.3)
144
145 plt.tight_layout()
146 plt.savefig('box_plots_distribution.png', dpi=300, bbox_inches='tight')
147 print("✓ Saved: box_plots_distribution.png")
148 plt.show()
149
150 # 2. Single Box Plot: Overall Enrollment Distribution
151 plt.figure(figsize=(10, 6))
152
153 # Prepare data for all age groups combined
154 all_ages_data = [
155     df['age_0_5'].dropna().values,
156     df['age_5_17'].dropna().values,
157     df['age_18_greater'].dropna().values,
158     df['Total_Enrollments'].dropna().values
159 ]
160
161 bp3 = plt.boxplot(all_ages_data, labels=['Age 0-5', 'Age 5-17', 'Age 18+', 'Total'],
162                  patch_artist=True, notch=True, showmeans=True,
163                  meanprops=dict(marker='D', markerfacecolor='red', markersize=8))
164
165 # Customize colors
166 box_colors = ['skyblue', 'lightcoral', 'lightgreen', 'gold']
167 for patch, color in zip(bp3['boxes'], box_colors):
168     patch.set_facecolor(color)
169     patch.set_alpha(0.7)
170
171 plt.xlabel('Enrollment Categories', fontsize=12, fontweight='bold')
172 plt.ylabel('Count', fontsize=12, fontweight='bold')
173 plt.title('Box Plot: Aadhar Enrollment Distribution Analysis\n(Shows Median, Quartiles, and Outliers)',
174          fontsize=14, fontweight='bold')
175 plt.grid(axis='y', alpha=0.3)
176
177 plt.tight_layout()
178 plt.savefig('box_plot_overall_analysis.png', dpi=300, bbox_inches='tight')
179 print("✓ Saved: box_plot_overall_analysis.png")
180 plt.show()

```

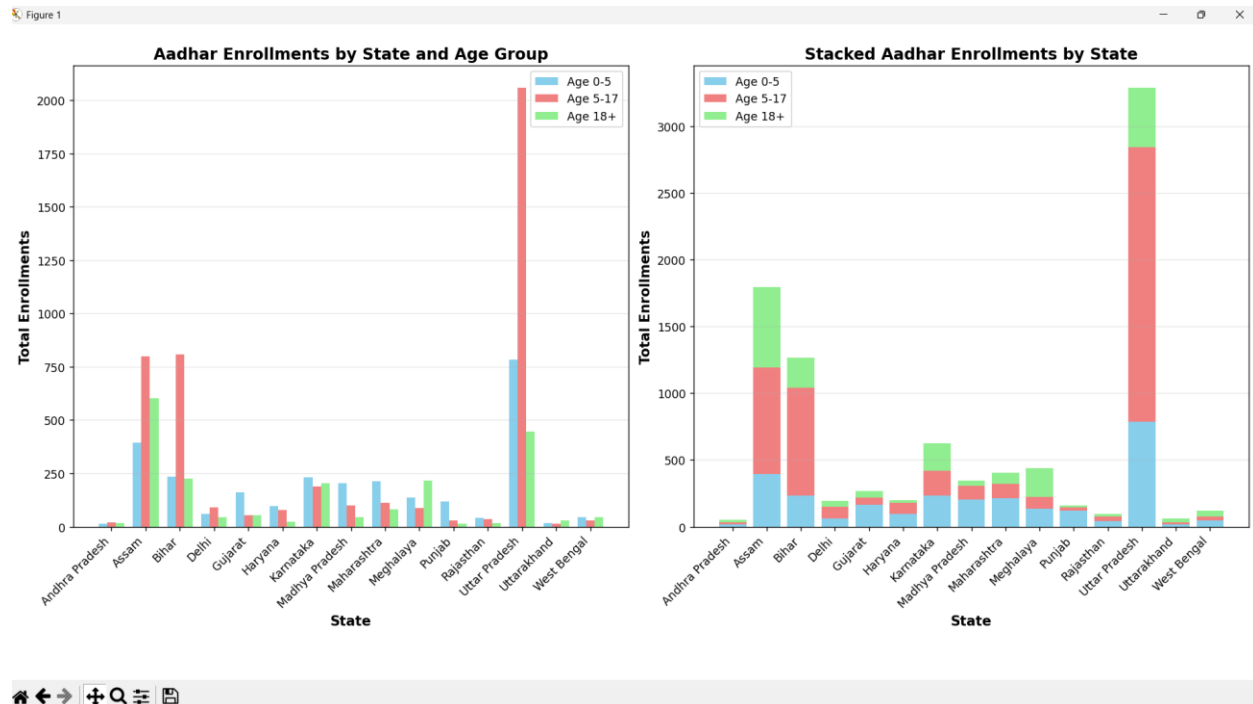
```

EXPLORER
EXP2
  .venv
  .vscode
  Image
  ~$P2 (Bar Chart ... U
  demo.py
  DOC.ipynb
  EXP2 (Bar Chart Visuali...
  EXP2.py 3
  Sample_Aadhar_Enroll...

EXP2.py > ...
181
182 print("\nBox plots displayed successfully!")
183 print("\nBox Plot Insights:")
184 print("- The box shows the interquartile range (IQR) - middle 50% of data")
185 print("- The line inside the box is the median")
186 print("- The diamond marker shows the mean")
187 print("- Whiskers extend to 1.5 IQR")
188 print("- Points beyond whiskers are potential outliers")
189
190 print("\n" + "="*50)
191 print("ALL PLOTS SAVED SUCCESSFULLY!")
192 print("="*50)
193 print("\nSaved Files:")
194 print("1. bar_charts_by_state.png")
195 print("2. top_10_districts_bar_chart.png")
196 print("3. box_plots_distribution.png")
197 print("4. box_plot_overall_analysis.png")
198 print("\nYou can now view all plots as image files in the current directory.")
199 print("Simply close each plot window as it appears to continue.")
200

```

Output



Chart's Visualizations Created:

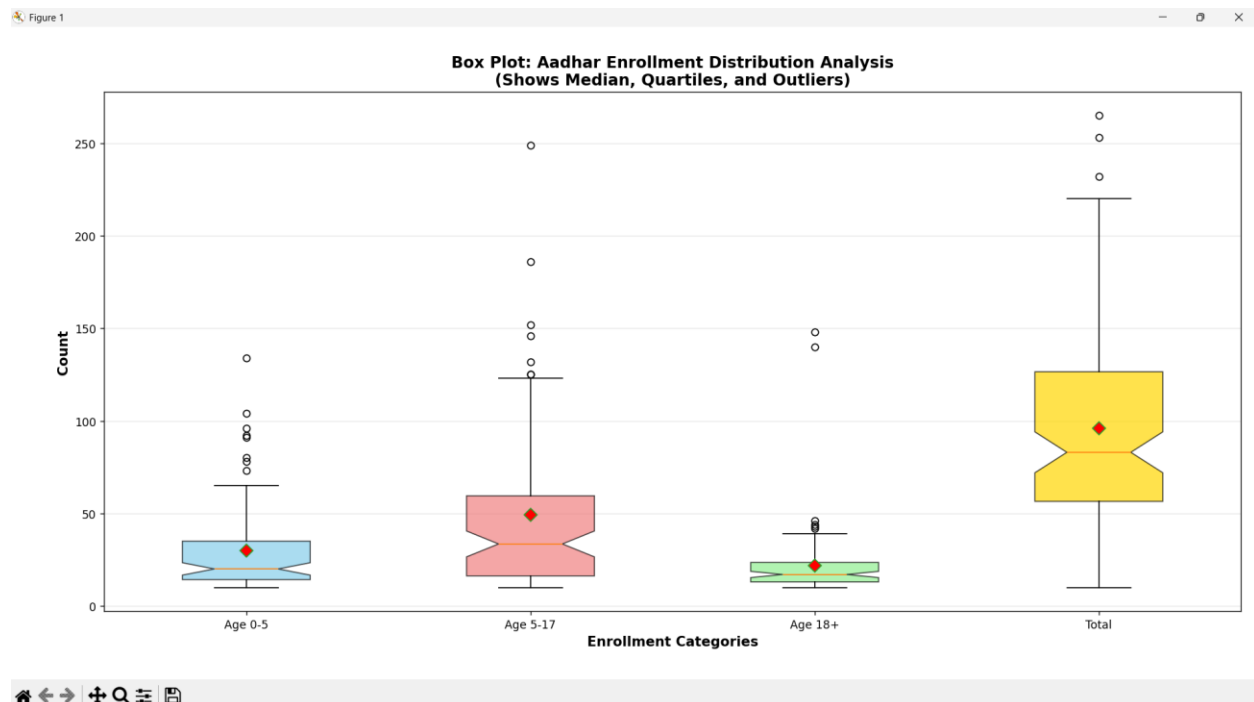
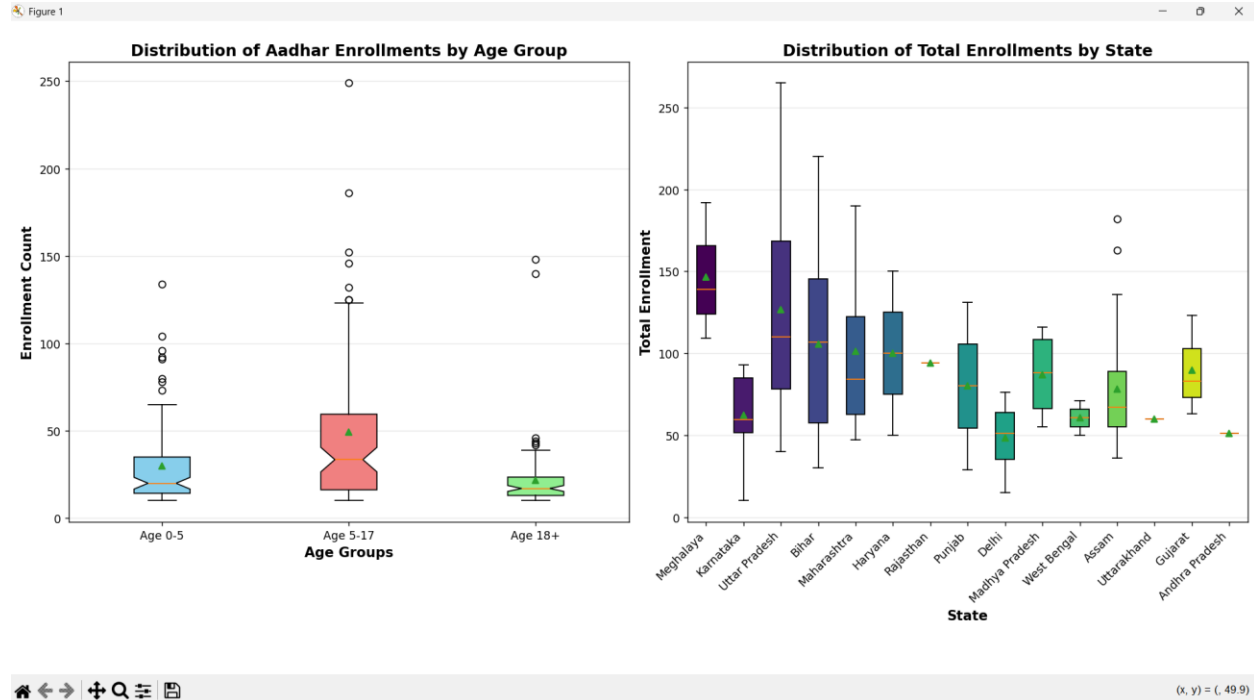
- Grouped Bar Chart** (Left panel)
 - Compares enrollments across different states
 - Shows three age groups side-by-side for each state
 - Uses different colors: Age 0-5 (blue), Age 5-17 (coral), Age 18+ (green)
- Stacked Bar Chart** (Right panel)
 - Shows total enrollments per state
 - Age groups stacked on top of each other
 - Gives a clear view of total enrollment volume per state

The bar charts are ideal for this dataset because they clearly compare categorical data (states/districts) with enrollment counts across different age groups, making it easy to identify patterns and regional differences in Aadhar enrollments.

The Bombay Salesian Society's DON BOSCO INSTITUTE OF TECHNOLOGY

(An Autonomous Institute affiliated to University of Mumbai)

Department of Information Technology



High Performing States

1) Uttar Pradesh

Highest overall enrollment in both graphs

- **Age 5–17** → Extremely high (largest among all states)
- **Age 0–5** → Very high
- **Age 18+** → Moderate but still large compared to many states

Uttar Pradesh is the **top performing state** with maximum Aadhaar enrollment, especially among school-age children.

2) Assam

Second highest performer

- High enrollment in **Age 5–17**
- High enrollment in **Age 18+**
- Good enrollment in **Age 0–5**

Assam shows **balanced and strong performance** across all age groups.

Equal Performing States

These states show **balanced but medium-level enrollments**.

1) Karnataka

- Almost equal enrollment in all age groups
- Moderate totals

Shows balanced Aadhaar awareness and participation.

Low Performing States

These states show **lowest Aadhaar enrollment totals**.

1) Andhra Pradesh

- Very low enrollments across all age groups

2) Uttarakhand

- Low enrollment in all age groups
 - Still falls in low performance group
-

Age Group Wise Performance Insight

Age 0–5 (Children)

- Highest → Uttar Pradesh
- Good → Assam, Bihar
- Lowest → Andhra Pradesh, Jharkhand

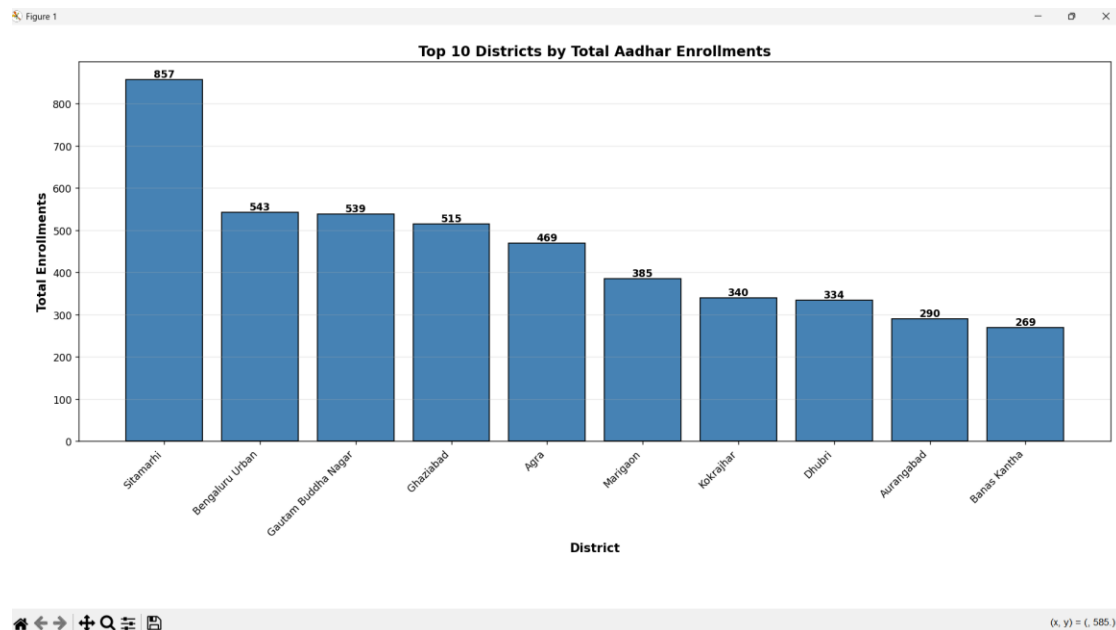
Age 5–17 (School Age)

- Highest → Uttar Pradesh (dominates strongly)
- Second → Assam & Bihar
- Lowest → Andhra Pradesh, Gujarat

Age 18+

- Highest → Assam
- Moderate → Uttar Pradesh
- Lowest → Andhra Pradesh, Jharkhand

TOP 10 Districts by Total Aadhar Enrollment



Conclusion

In this experiment, data visualization was successfully performed on the Aadhar Enrollment dataset. A new Total_Enrollment column was derived by summing the three age-group columns. Grouped and stacked bar charts revealed the distribution of enrollments across age groups for each state, highlighting which states had higher participation in specific age brackets. The Top 10 districts bar chart with value labels made it easy to identify the highest-enrolling districts at a glance. Box plots provided insight into the spread, median, IQR, and outliers of enrollment counts across age groups and states. Together, these visualizations offered a comprehensive and interpretable picture of enrollment patterns across India's Aadhar dataset.